

Password Manager User Documentation

Introduction

Welcome to the Password Manager! This tool helps you securely store and manage your passwords using state-of-the-art encryption. With features like user authentication, OTP verification, and comprehensive password management, this password manager ensures your sensitive information is protected, and can be retrieved by the user anytime.

Installation

To install the Password Manager, follow these steps:

- Ensure you have Python installed on your system. You can download it from python.org.
- Download the `Final\_Password\_Manager.py` and `Final\_UI.py` script to your preferred directory. Run `Final\_UI.py`.

Setup and Configuration

Before you start using the password manager, a few setup steps are required:

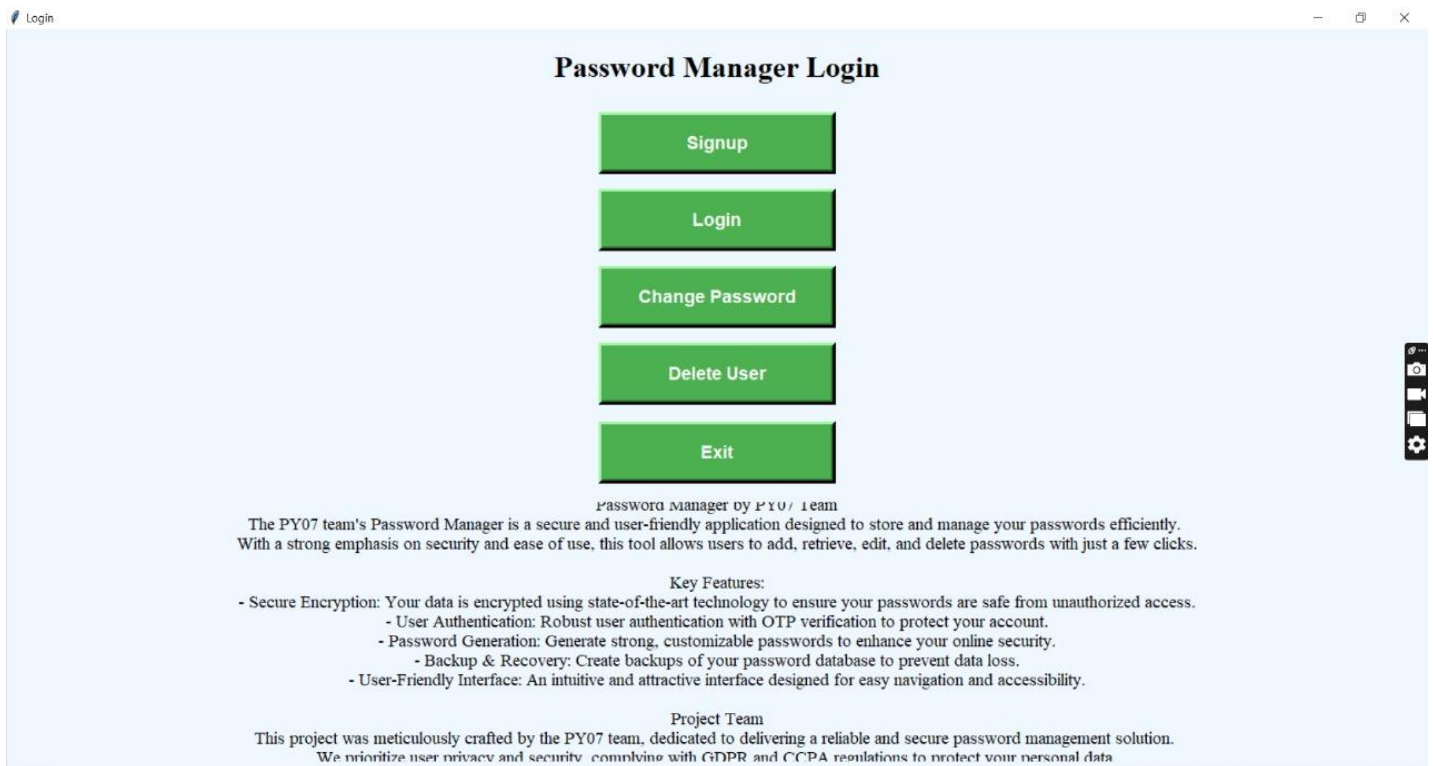
1. Generate Encryption Key:

The script automatically generates and saves an encryption key if it doesn't exist. This key is used for encrypting and decrypting your data.

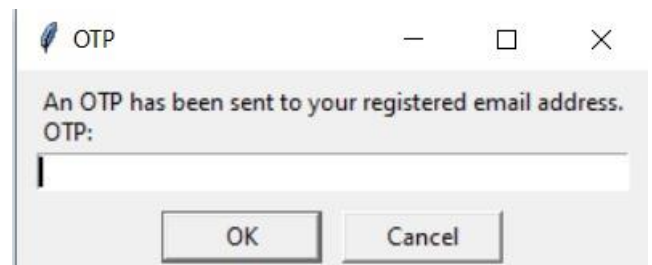
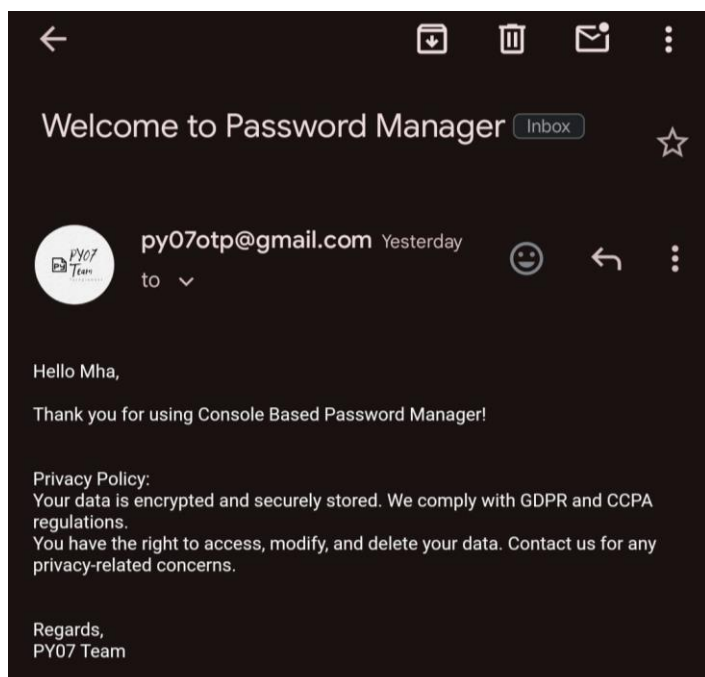
2. Database Initialization:

The script initializes two SQLite databases: `password\_manager.db` for storing passwords and `Account.db` for user account management.

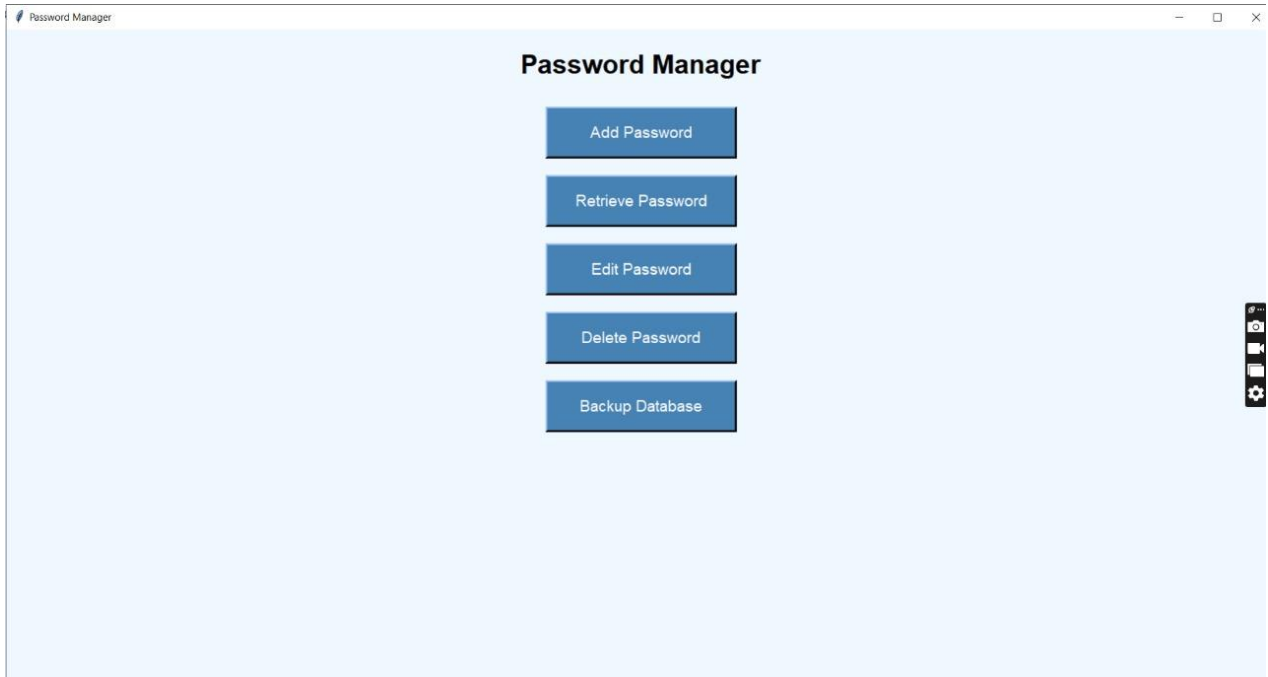
Using the Password Manager



This is the homepage of the Password Manager Application. You have the option to signup to create your own account, login to your existing account. You can also change password of your account or delete your account as well. A welcome message is sent to your mail ID if you're a first time user. To login, give in your credentials, such as username and password. It is also further backed up by 2FA (two factor authentication) in the form of OTP sent to your registered mail ID as shown below.

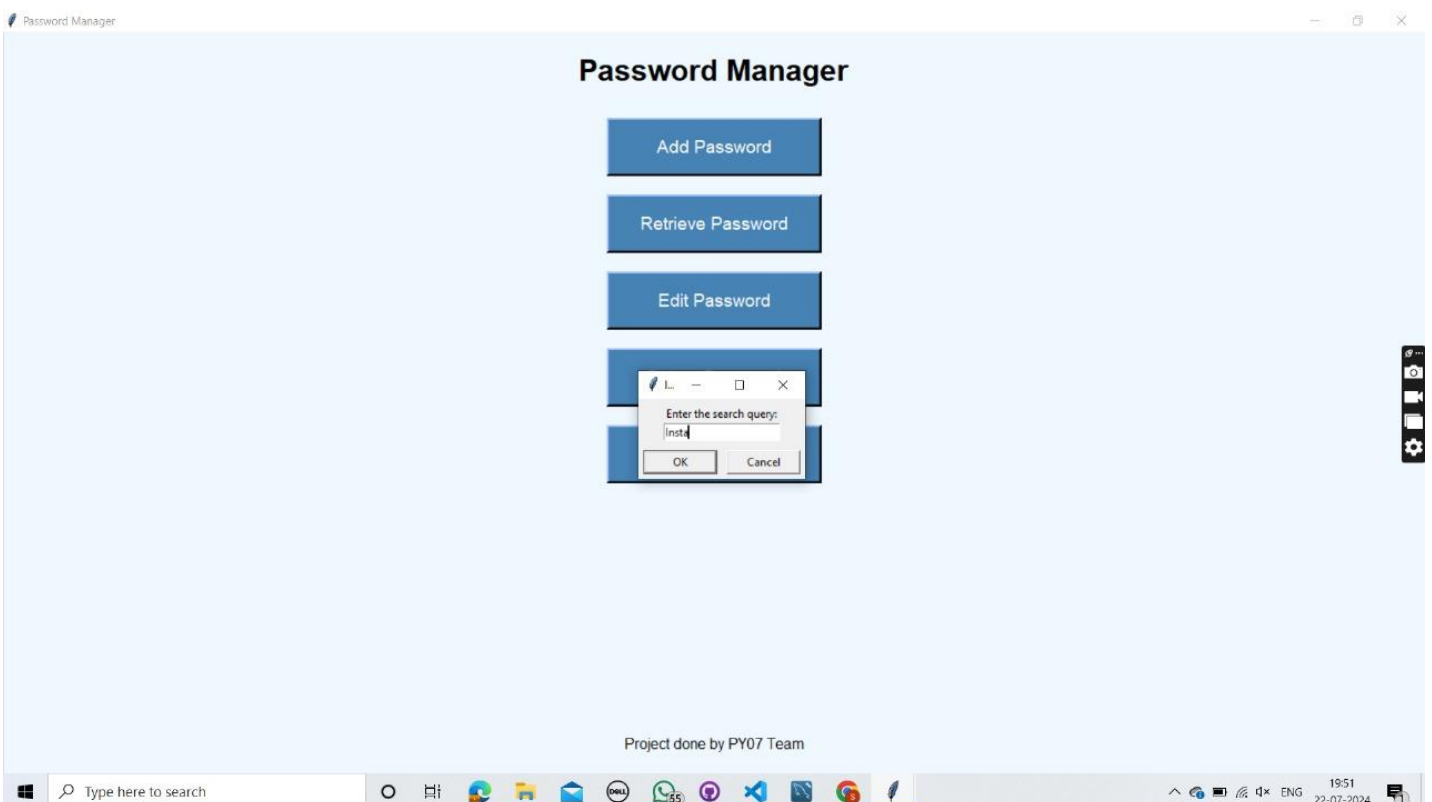
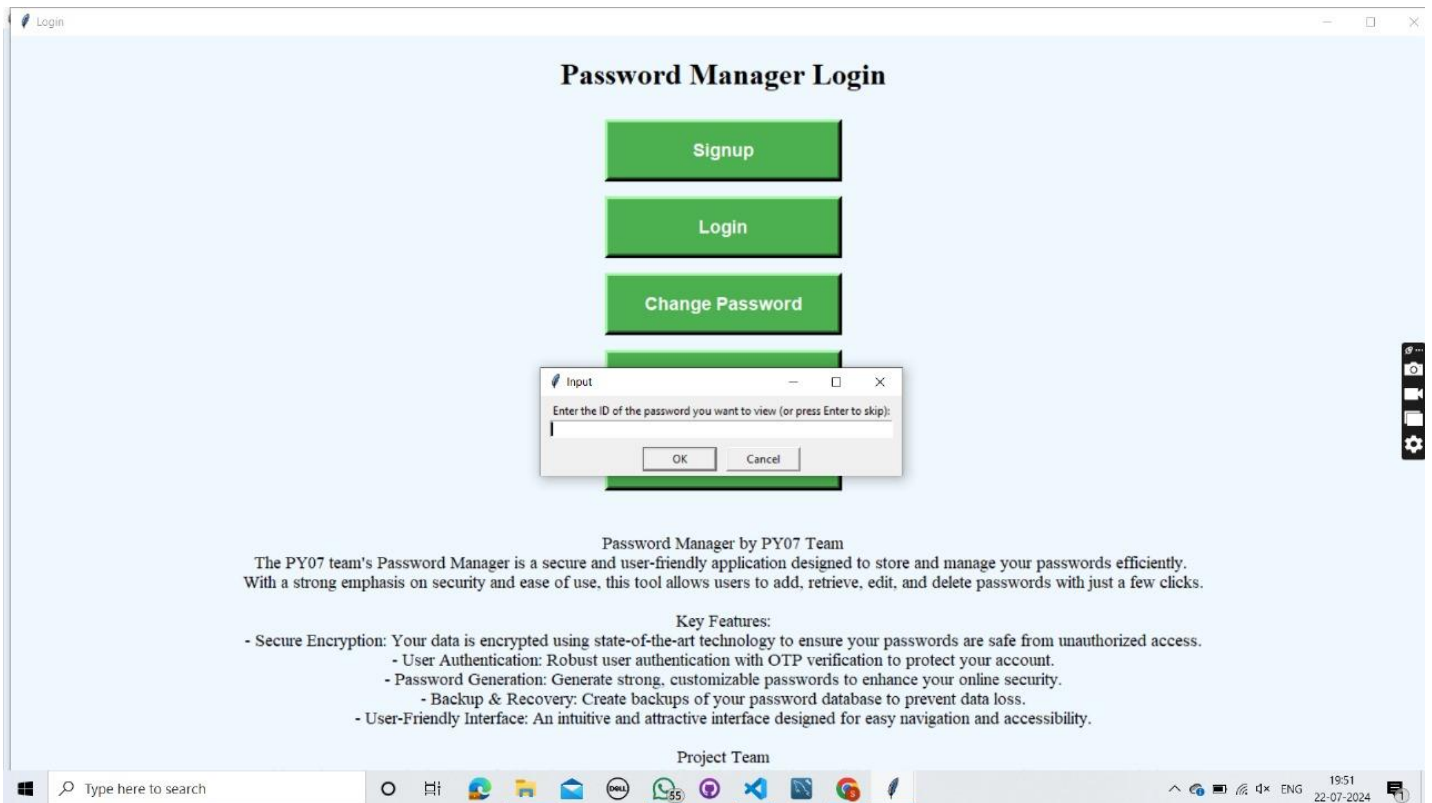


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Once you have created or logged in your account, you have 5 options presented to you. You can add password to a desired application, retrieve existing passwords, edit or delete password of any application you've given, and finally back it up in the database. You can either generate a password using our inbuilt Password Generator or enter your own. The passwords you enter are encrypted using industry-standard encryption algorithms, such as AES (Advanced Encryption Standard).

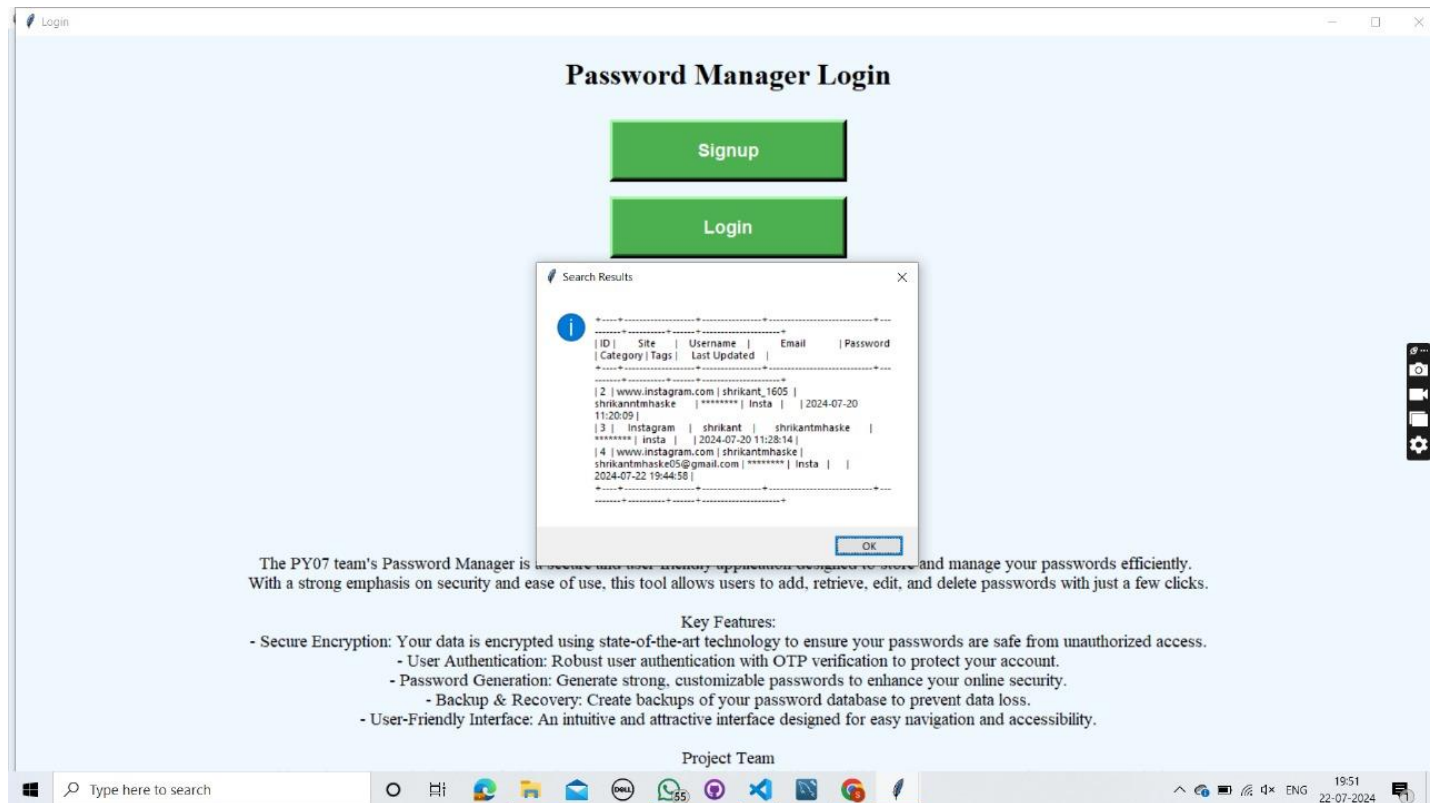
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Once you have added password of a particular application, you can retrieve/view the password by entering the name of the application whose password you want to retrieve, and the application will retrieve it for you. You can also edit the password, but to edit you must give the OTP first, then only will you be allowed to edit the existing password. This is to comply with the security practices to keep all passwords very safe and secure.

Backup and Restore



You can backup and restore your passwords for different applications in our secure database and synchronize it across multiple devices. It displays the application name, username, password (encrypted) and the data and time it was entered/last updated.

Features of the Password Manager Application:

- An intuitive and user-friendly interface, easy to navigate and provides a seamless experience for users to manage their passwords.
- A secure authentication mechanism to ensure that only authorized users can access the password manager, backed up with a layer of two-factor authentication (2FA).

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- A strong encryption mechanism to securely store passwords, utilizing algorithms such as AES (Advanced Encryption Standard), to encrypt and decrypt passwords.
- An inbuilt Password Generator to generate strong and unique passwords where users have the option to specify password length, complexity, and include a mix of uppercase and lowercase letters, numbers, and special characters.
- A secure storage mechanism to store user passwords, utilizing secure databases to securely store and manage the passwords.
- A robust Password Organization system wherein users can categorize and organize their passwords using categories to group passwords based on the type of account or application.
- Users can also search for specific passwords based on account names, keywords, or categories as well as have the ability to back up their password data and synchronize it.
- Additional security measures, such as password strength indicators, session timeouts, and password reuse detection have also been implemented.
- We comply with relevant data protection regulations, such as GDPR (General Data Protection Regulation) or CCPA (California Consumer Privacy Act).
- Thorough testing, including unit testing, integration testing, and security testing have been done to ensure the reliability and security of the password manager. We also promise regular software updates and patches to address any discovered vulnerabilities.

Troubleshooting and FAQs

➤ Common Issues

1. Unable to receive OTP:

- Check your spam folder.
- Ensure your email server settings allows receiving emails from the sender.

2. Login Failure:

- Ensure you are entering the correct email along with username and password.
- If you forget your password, you can re-enter a new password for a given account, which is strengthened by two-factor authentication (OTP).

3. Database Errors:

- Ensure the database files are not corrupted and have appropriate read/write permissions.

FAQs

- **Why should I use a password manager?**
Password managers help you create and store strong, unique passwords for different accounts, enhancing your security and saving you from the hassle of remembering multiple passwords.
- **Is my data secure?**
Yes, the password manager uses industry-standard encryption to protect your data. Your passwords are stored in an encrypted format that can only be decrypted by you.
- **How do I generate a strong password?**
Use the built-in password generation feature, which allows you to specify the length and complexity (including uppercase, lowercase, numbers, and special characters) of your password.
- **How do I search for a specific password?**
Use the search functionality to find passwords by account names, keywords, or categories.
- **How does the password manager ensure the security of my passwords?**
The password manager uses AES encryption to securely store your passwords. It also includes features like session timeouts and password strength indicators to enhance security.
- **What is two-factor authentication (2FA), and how do I use it?**
Two-factor authentication adds an extra layer of security by requiring a second form of verification in addition to your password. You have the option of getting an OTP whenever you want to login.
- **What happens if I forget my master password?**
If you forget your master password, you can use the account recovery feature to reset it. This may involve answering security questions or using a recovery email.
- **How do I back up my passwords?**
You can back up your password data using the backup feature, which allows you to store your data securely in our database.
- **Why can't I log in to my account?**
Ensure you are entering the correct username and password. If you have forgotten your password, use the account recovery feature.
- **Why is my session timing out frequently?**
For security reasons, sessions are set to timeout after a period of inactivity. You can adjust the session timeout settings if needed.
- **What are the best practices for creating passwords?**
Use unique passwords for each account, avoid common words or easily guessable information, and use the password generation feature to create strong passwords.

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Thank you for using the Password Manager. We hope this tool helps you keep your passwords secure and organized. If you have any questions, feedback, or need further assistance, please do not hesitate to contact our support team. Your security is our priority, and we are committed to providing you with the best possible experience.

Stay safe and secure!

By Py07,

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